Lecture 11, Math 441

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(cont.)

We are working to prove the following theorem:

Given the IVP

$$\begin{cases} y' = f(x, y) \\ \\ y(0) = 0 \end{cases}$$

If $\exists$ a rectangle R, $|x| < a$ and $|y| < b$, where f and f_y are continuous then $\exists$ an interval $|x| < h$ where $\exists$ one and only one solution to the IVP.

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We rewrote as an integral equation: $y(x) = \int_0^x f(s, y(s))\,\mathrm{d}s$

and we defined a sequence of functions:

$y_0(x) = 0; \qquad y_n(x) = \int_0^x f(s, y_{n-1}(s))\,\mathrm{d}s$

We also defined M to be the maximum value of $|f|$ in R, and T to be the maximum of $|f_y|$ in R.

We then showed that all of the $y_n(x)$ exists and that the sequence $\{y_n\}$ is uniformly convergent in $|x| \leq h$ for some h.

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Let's call the limit $y(x)$: $\{y_n(x)\} \to y(x)$

To prove that $y(x)$ satisfies the equation, we just take the limit of

the definition of the sequence:

$\lim_{n\to\infty} y_n(x) = \lim_{n\to\infty} \int_0^x f(s, y_{n-1}(s))\,\mathrm{d}s$

which gives $\quad y(x) = \int_0^x f(s, y(s))\,\mathrm{d}s,$

i.e. $y(x)$ satisfies the equation and $y(0) = 0$.

(We can take the limit inside the integral and the function f

because the convergence is uniform and f is continuous.)

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To prove uniqueness, let's assume there are two solutions:

$\Phi(x) = \int_0^x f(s, \Phi(s))\,\mathrm{d}s;\ \ \Phi(0) = 0$ and

$\Psi(x) = \int_0^x f(s, \Psi(s))\,\mathrm{d}s;\ \ \Psi(0) = 0$

By definition: $|\Phi(x) - \Psi(x)| = |\int_0^x [f(s, \Phi(s)) - f(s, \Psi(s))]\,\mathrm{d}s| \leq$

$\int_0^x |f(s, \Phi(s)) - f(s, \Psi(s))|\mathrm{d}s \leq T \int_0^x |\Phi(s) - \Psi(s)|\mathrm{d}s$

using an inequality we found last time.

Let's define $U(x) = \int_0^x |\Phi(s) - \Psi(s)| \mathrm{d}s$ (which is ≥ 0 if $x > 0$)

Note that $U(0) = 0$ and $U'(x) = |\Phi(x) - \Psi(x)|$

The result in the above slide means that $U'(x) \leq T\, U(x)$ or

$U'(x) - T\, U(x) \leq 0$

Multiplying by the positive quantity e^{-Tx} we get

$U'(x)\, \mathrm{e}^{-Tx} - T\, U(x)\, \mathrm{e}^{-Tx} \leq 0 \;\; \rightarrow \;\; \left(U(x)\, \mathrm{e}^{-Tx}\right)' \leq 0$

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Now $U(x)\,\mathrm{e}^{-Tx} = 0$ when $x = 0$

We already know that $\left(U(x)\,\mathrm{e}^{-Tx}\right)' \leq 0$ which means $U(x)\,\mathrm{e}^{-Tx}$

cannot grow, so it cannot become positive. This means that

$U(x)\,\mathrm{e}^{-Tx} \leq \rightarrow \quad U(x) \leq 0$

Given that $U(x) \geq 0$ it must be that $U(x) = 0$, i.e.

$\int_0^x |\Phi(s) - \Psi(s)|\mathrm{d}s = 0 \rightarrow |\Phi(s) - \Psi(s)| = 0 \;\forall s \in (0, x) \rightarrow$

$\Phi(x) = \Psi(x)$, i.e. the solution is unique.

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Ex: The IVP: $y' = 2(y+1)$; $y(0) = 0$ can be solved easily as the equation is separable. But let's practice using successive approximations to find the solution.

Rewrite as an integral equation: $y(x) = \int_0^x 2(y(s)+1)\,\mathrm{d}s$

and start the sequence: $y_0(x) = 0$

$y_1(x) = \int_0^x 2(y_0(s)+1)\,\mathrm{d}s = \int_0^x 2\,\mathrm{d}s = 2x$

$y_2(x) = \int_0^x 2(y_1(s)+1)\,\mathrm{d}s = \int_0^x 2(2s+1)\mathrm{d}s = 2x^2 + 2x$

$$y_3(x) = \int_0^x 2(y_2(s) + 1)\,\mathrm{d}s = \int_0^x 2(2s^2 + 2s + 1)\,\mathrm{d}s = \frac{4}{3}x^3 + 2x^2 + 2x$$

So far it seems that

$$y_1(x) = (2x)$$

$$y_2(x) = 2x^2 + 2x = \frac{4}{2}x^2 + 2x = \frac{(2x)^2}{2} + (2x)$$

$$y_3(x) = \frac{4}{3}x^3 + 2x^2 + 2x = \frac{8}{6}x^3 + \frac{4}{2}x^2 + 2x = \frac{(2x)^3}{3!} + \frac{(2x)^2}{2!} + (2x)$$

and we can show that $y_4(x) = \frac{(2x)^4}{4!} + \frac{(2x)^3}{3!} + \frac{(2x)^2}{2!} + (2x)$

So we can expect that

$$y_n(x) = 2x + \frac{(2x)^2}{2!} + \cdots + \frac{(2x)^n}{n!} = \sum_{k=1}^{n} \frac{(2x)^k}{k!}$$

Let's check that y_{n+1} follows this formula so that we can prove it

by induction.

$$y_{n+1}(x) = \int_0^x 2\left(1 + \sum_{k=1}^{n} \frac{(2x)^k}{k!}\right) \mathrm{d}s = 2x + 2\sum_{k=1}^{n} \frac{(2)^k (x)^{k+1}}{(k+1)k!} =$$

$$2x + \sum_{k=1}^{n} \frac{(2)^{k+1}(x)^{k+1}}{(k+1)!} = 2x + \sum_{k=2}^{n+1} \frac{(2x)^k}{k!} = \sum_{k=1}^{n+1} \frac{(2x)^k}{k!}$$

so we know we have the right formula.

Now let's rewrite y_n as

$$y_n(x) = \sum_{k=1}^{n} \frac{(2x)^k}{k!} = -1 + \sum_{k=0}^{n} \frac{(2x)^k}{k!}$$

We can now take the limits for $n \to \infty$. The sum converges to e^{2x}.

Therefore the solution to the IVP is

$$y(x) = -1 + \mathrm{e}^{2x}$$

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